DATA MINING PROJECT

Master’s in data science and advanced Analytics

NOVA Information Management School

Universidade Nova de Lisboa

Amazing International Airlines Inc.

Group 12

Luis Mendes, 20221949

Margarida Ourives, 20221809

Simon Sazonov, 20221689

Veronica Mendes, 20221945

Fall/Spring Semester 2025-2026

TABLE OF CONTENTS

1. ABSTRACT

This report presents the Exploratory Data Analysis (EDA) for the Amazing International Airlines Inc. (AIAI) customer segmentation project, aligning with the Business and Data Understanding phase of the CRISP-DM methodology. The study explores the customer loyalty membership data and corresponding flight activity collected over a three-year period to identify initial patterns for Value-based, Behavioral, and Demographic segmentation. Data quality checks confirmed consistent structure and minimal missing values. A few invalid or duplicate records were removed; visualizations and new variables were created. The analysis revealed compelling initial trends, indicating varying levels of customer engagement, spending patterns, and travel preferences across the dataset. These insights will serve as critical input for the next phase and the development of robust customer clusters.

2. INTRODUCTION

Amazing International Airlines Inc. (AIAI) faces the challenge of designing personalized marketing and services to stay competitive in the airline industry. The primary goal of this project is to develop a data-driven customer segmentation strategy for AIAI's loyalty program members using the customer base and three years of flight activity data. The segmentation will be approached from three critical, holistic perspectives: Value-based (economic contribution), Behavioral (purchasing habits and frequency), and Demographic (attribute patterns) to provide a complete understanding of the customer base for strategic targeting. This report focuses on the Exploratory Data Analysis (EDA) and the creation of new features, which establishes a foundational understanding of the data, its quality, and the relationships among key variables. These findings will guide the upcoming data preparation, modeling (clustering) and evaluation phase.

3. DATA ANALYSIS

The Data Analysis phase involved analyzing the two datasets (Customer and Flights) to assess structure, completeness, and initial patterns.

3.1 Customer Dataset

The Customer Dataset originally contained 16.921 observations, representing all customers from Canada. It includes Loyalty#, First Name, Last Name, Customer Name, Country, State, City, Latitude, Longitude, Postal code, Gender (male, female), Education, Location code (suburban, rural, urban), Income, Marital status (Married, single, divorced), Loyalty status (Satar, Nova, Aurora), EnrollmentDateOpening, CancellationDate, Customer Lifetime Value, EnrollmentType (standard, 2021 promotion).

Missing values (evaluated column by column): CancellationDate (14.611 records, 86% missing): This high rate is not considered a data just that as most customers have not officially left the program. Income and Customer Lifetime Value (CLV) (for both: 20 records, 0.11%): Records missing in both were deleted as they were associated with customers who canceled the subscription on the same day, representing non-viable program members. Two customers enrolled and canceled on the same, since they don't have any real records of flights we drop these customers.

Duplicate Removal: A total of 327 records (1.93%) with identical Loyalty#, but differing characteristics were removed to ensure a unique key for each customer, a requirement for integration.
As a result, 16,572 customers (removed 349 records, 2,06% of the hole dataset).

3.2 Flight Dataset

The Flight Database contained 608.436 flight records with no missing values and multiple entries per customer. It includes Loyalty#, Year, Month, YearMonthDate, NumFlights, DistanceKM, NumFlightsWithCompanions, PointsAccumulated, PointsRedeemed, DollarCostPointsRedeemed.

Overall, 5.778 duplicate records were removed, keeping one instance per duplicate. To handle repeated monthly entries, 5.996 customer–month records were aggregated, summing key activity variables, to create a single consolidated row per month. A further 301.621 zero-flight records were removed as they contained no activity, along with about 5.000 records (2%) showing flight distances without corresponding counts, as the lack of route information prevented accurate estimation.

The cleaned dataset includes 15.101 customers with flight data and 1.471 loyalty members without travel records, covering activity from 01-01-2019 to 01-12-2021. Based on the analysis we concluded that DistanceKM was used to estimate the PointsAccumulated ($PointsAccumulates = \frac{DistanceKM}{10}$) and PointsAccumulated to estimate the DollarCostPoints ($DollarCostPoints = \frac{Points}{100}$).

3.3 Merging Datasets

For both datasets, since we do not have a 1:1 correspondence, to join these datasets we will need to perform some aggregations on the flight dataset to have only 1 value per customer), we performed some total, yearly and monthly aggregations.

3.3.1 New Features

We create some variables for customers: Re-engaged people (Indicates whether a customer rejoined the loyalty program after a previous cancellation), Days with us (Represents the total number of days a customer stayed enrolled in the loyalty program), Days until reengagement (Measures how long it took for a previously canceled customer to rejoin the program). For flights: Recency (measures the time elapsed since the last flight acquisition) , Perc Flights with Companions ($\%\ Comp = \frac{Number\ of\ flights\ with\ companions}{Number\ of\ flights} \times 100$), Dollar Cost Points Remaining (unredeemed Dollar value held by the customer in the form of loyalty points). We are no longer going to use variables: point accumulation, point redeemed and number of flights with comparison because these variables can be calculated can be calculated using the new ones.

3.4 Exploratory Data Visualization

Exploratory visualizations were created to understand the structure and variability of the datasets.

3.4.1 Customers Data Frame

The categorical variables in customer dataset are balanced; people came into the program in a uniform way, the Loyalty status that most people have is Star, followed by Nova and lastly Aurora, the proportion of the characteristics of the clients is balance, meaning that the loyalty status don't change the demographic aspect of the clients See Annex1.

We observe a uniform data enrollment of the customers, in contrast the cancelation date has been increasing over time, and we need to address this problem and see what the customers don't like to reduce the increasing cancelation trend. Other variables are going to be analyzed Value based, behavioral, demographic section.

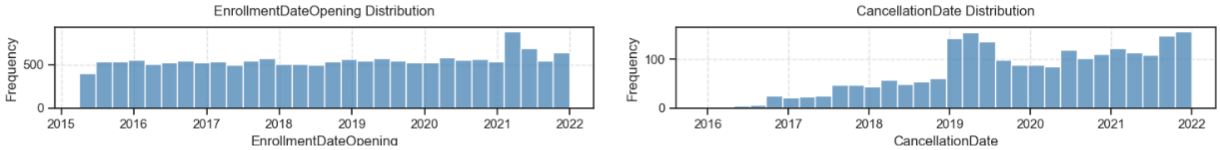


Figure 1 - Enrollment and Cancellation date

3.4.2 Flights Data Frame

The flight dataset has a clear growing trend, as shown by the increase in yearly totals. However, a strong seasonal pattern is evident, travel demand concentrated in the summer months (June-August) and Christmas holidays and falling to their lowest points in the beginning of the year (January-February).

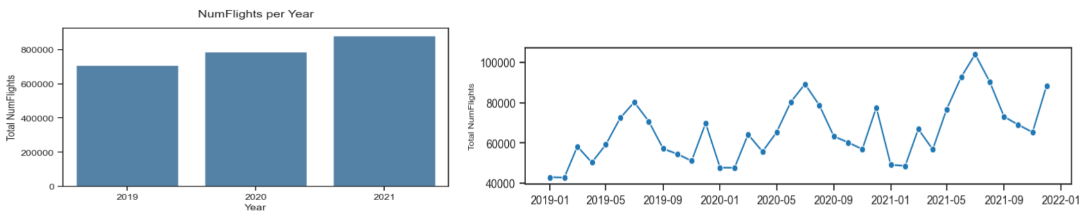


Figure 2- Evolution over time

3.5 Cluster Variables

Using the old features, new features and aggregation we group our variables based on Value of the customer (grouping customers according to their economic contribution), demographics (categorizing customers by age, occupation, or other attributes to reveal different interaction patterns) and Habits/Behavioral (purchasing habits and travel behaviors). See Annex2.

3.5.1 Value-based

There are two groups of Re-engaged customers: the ones that enter on the same day in the 2021 promotion, and customers who returned after approximately 850, maybe for a special campaign program that activated after a long inactivity period. See Annex3.

Customer Lifetime Value shows a strong right-skewed distribution. Most customers contribute a moderate lifetime value, while a smaller group of outliers represents the company's most profitable and loyal members. This pattern indicates that a small segment of customers generates a large portion of total revenue. See Annex4.

The Relationship between total distance and the number of flights exhibits a high degree of positive correlation; this outcome is intuitively logical, as an increased frequency of flights directly contributes to a greater accumulated distance. A notable observation concerning customer behavior is the low rate of points redemption, suggesting that a significant portion of the customer base retains their accumulated loyalty points rather than utilizing them for travel benefits. See Annex5.

It indicates a predominant trend of recent purchasing activity among the customer base, as the majority have made a purchase within the last 100 days. However, a smaller, distinct segment demonstrates long-term inactivity, with some customers having not transacted for over two years. Annex6.

3.5.2 Behavioral

As we saw in Value-based, yearly and monthly features show clear and logical trends, the total distance is very positively correlated with the number of flights and points accumulated. Critically, many customers do not redeem their points, and a minority segment possesses zero flight records.

The Number of Flights, Total Distance, Companion Percentage, and Dollar Cost of Remaining Points exhibit a perfect positive correlation between 2019 and 2020. Since these years convey identical information, only one year should be selected for subsequent analysis to mitigate redundancy and multicollinearity, Annex7. Travel companion incidence varies widely among customers (from 0% to 100%), fluctuating annually (showing the same on monthly data). The 2021 data constitute an isolated anomaly, generating a pattern with no correlation to historical trends, suggesting that a specific event or action in that year altered typical behavior, Annex8.

3.5.3 Demographic

We have the same number of males and females, they have a balanced class distribution of education, Location code (same number per class) and marital status (having more customers married, them singles and a few divorced). We have more people on bachelors, then college and a few on the rest categories of the education variable (master, Phd and high school or below), but they have the same proportion of location code. What they don't have the same proportion is the marital status: college customers the majority are single, the masters are divorced or married, having the rest categories (bachelor, doctor/Phd and high school or below) more or less similar proportions. See Annex9.
The income distribution is heavily skewed because customers identified with a college education show a large number of zero-income values, unlike other education levels such as high school, master's, and doctorate, which follow more reasonable patterns, with the bachelor's group the one with higher income among all groups. Annex10.

3.5.4 Geographic Distribution

This analysis covers 11 Canadian Provinces/States and 29 cities, Annex11. All customers reside in the geographical center of the city; however, their specific classifications: Urban, Suburban, or Rural, are varied. Each city has different numbers of customers, Toronto, Vancouver and Montreal are the higher ones with more than 2000 customers approximately and are near popular airports, See Annex12. The mean customer lifetime value demonstrates near-uniformity across cities, centering around 4.000$. Notable deviations include Charlottetown (4.900$), whose island status logically supports increased air travel expenditure, and the highly remote northern city of Whitehorse (4.500$), See Annex13. The city of Kingston exhibits the greatest depletion rate of points, indicated by a negative average residual dollar cost. Additionally, elevated points usage is observed among travelers located near big airports, See Annex14. Based on an average transaction inactivity period exceeding 300 days, service provision to Sudbury, Vancouver, and Fredericton is presumed to be suspended. The status of Vancouver warrants specific examination, considering its role as a major airport hub, See Annex15. There is no clear distinction in demographic terms in relation to the cities; the characteristics of the clients are balanced in relation to the cities.

4. RESULTS

The exploration analysis showed that the AIAI datasets are well structured and provide consistent, useful information for understanding customer behavior. After data cleaning, 16.572 valid customer records remained in the Customer Dataset, and 15.101 customers have correspondence with flight data in the Flights Dataset.

While a simple analysis of the relation between two or three variables doesn't yield to distinct groups, their combination may be key to revealing underlying patterns tied to value-based customer contributions. Since each variable offers unique insight, we are going to keep them and employ a cluster analysis (on the second part of the project) that will be essential to effectively understand the interaction of these metrics and accurately segment customers based on their economic value to the company.

The Behavioral dimension revealed group of customers based on the Percentage of Flights with Companions (either 0% or 100%) and the Dollar Cost of Points Remaining (either used or unused). Given the large number of associated variables, the utility of remaining features will be confirmed during clustering. We anticipate that a dimensionality reduction technique, such as Principal Component Analysis (PCA) or Autoencoders, may be beneficial for generating more robust and well-formed clusters, since there are too many variables. Furthermore, a high degree of collinearity was observed between the annual data for 2019 and 2020, necessitating a cautious approach to their inclusion.

Demographic features, including Education and Income, appear to be highly significant differentiating factors for segment formation. This segmentation potential is further amplified when integrated with Geographical variables. A clear disparity in customer behavior across different cities was identified, thus justifying the critical decision to retain Longitude and Latitude as essential variables for subsequent customer segmentation efforts.

Overall, the results demonstrate that the data is suitable for segmentation and that clear differentiating factors exist across value, behavioral, demographic, and geographic dimensions. A self-service dashboard was created to analyze these variables: https://aiai-dashboard-dm.streamlit.app/

5. CONCLUSION

This Exploratory Data Analysis (EDA) successfully established a solid foundation for the Amazing International Airlines Inc. (AIAI) customer segmentation project. The datasets are clean, consistent, and informative. Data quality issues such as missing values and duplicates were resolved, ensuring a reliable clear pattern across Value-based, Behavioral, and Demographic dimensions, confirming their suitability base for subsequent modeling.

The next phase will apply clustering techniques to group customers according to these patterns. Feature selection will emphasize economic contribution and flight behavior, complemented by demographic and spatial context. This integrated approach aims to create a comprehensive understanding of their customers, so that on the second phase of the project it could be possible to define a final segmentation framework that supports AIAI in crafting marketing strategy.

6. ANNEXES

Annex AI Usage Statement

AI tools (ChatGPT) were utilized for support functions, namely generating code comments, refining sentence structure, proofreading text, and assisting with initial parameters for visualizations. All insights, conclusions, and recommendations are the result of our independent analysis and critical thinking.

Annex Contribution Statement

Individual student contributions to deliverables.

Luis Mendes (Student ID: 20221949)

- Notebooks Code, visualizations and interpretation of the code;
- Geographic Code;
- Revision of report;
- Dashboard.

Margarida Ourives (Student ID: 20221809)

- Report;
- Poster.

Simon Sazonov (Student ID: 20221689)

- EDA graphics and interpretation;
- Dashboard.

Veronica Mendes (Student ID: 20221945)

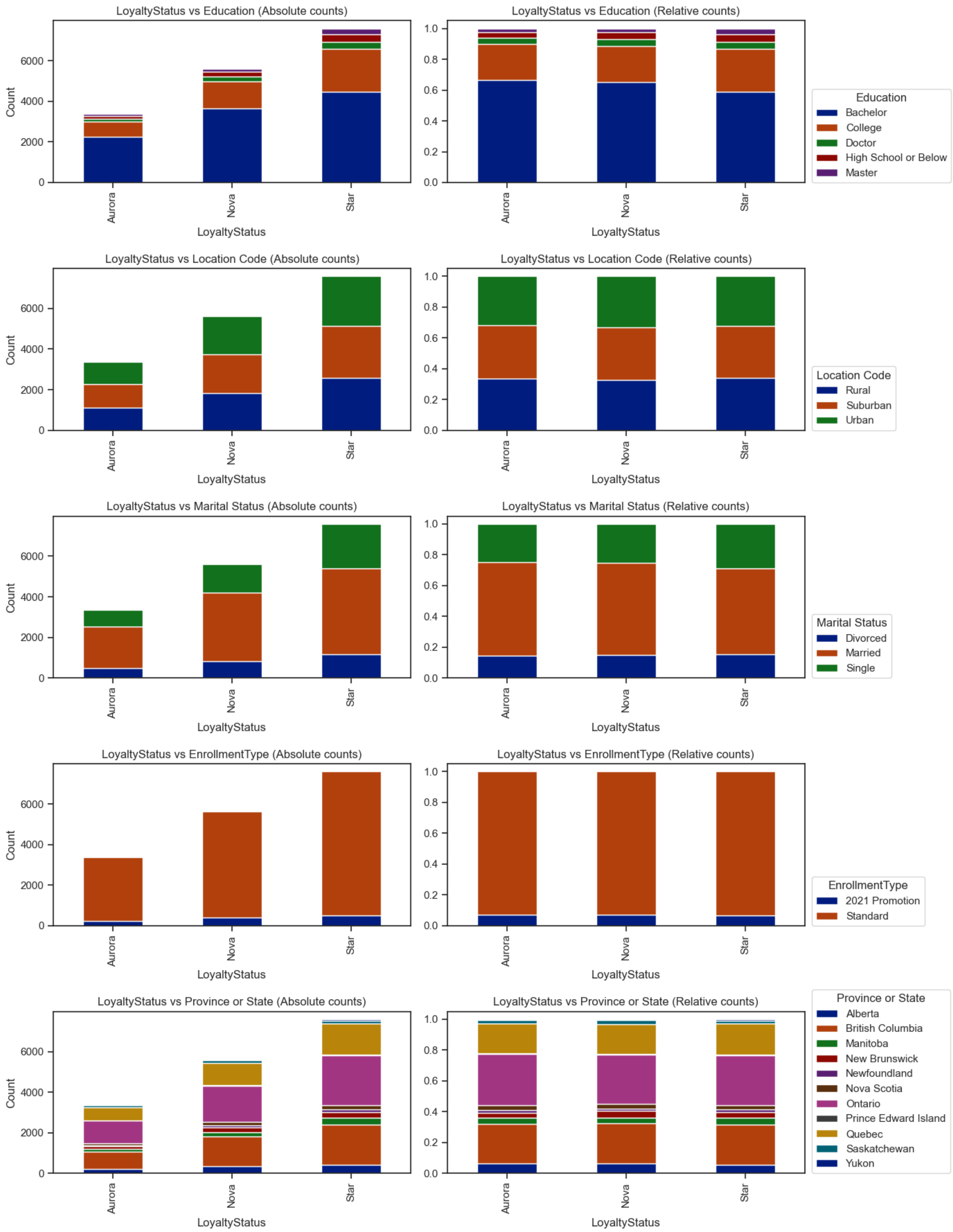
- Initial exploration of customers and flights dataset;
- Report;
- Poster.

All members contributed to collaborative discussions and ideation.

Annex Responsibility Statement

We, Group 12, certify that this report represents our original analytical work and interpretations. While AI tools were utilized as specified above. We collectively take full responsibility for the accuracy, quality, and integrity of this submission.

Annex1:


LoyaltyStatus vs Education (Absolute counts)
LoyaltyStatus vs Education (Relative counts)
Education
Bachelor
College
Doctor
High School or Below
Master
LoyaltyStatus vs Location Code (Absolute counts)
LoyaltyStatus vs Location Code (Relative counts)
Location Code
Rural
Suburban
Urban
LoyaltyStatus vs Marital Status (Absolute counts)
LoyaltyStatus vs Marital Status (Relative counts)
Marital Status
Divorced
Married
Single
LoyaltyStatus vs EnrollmentType (Absolute counts)
LoyaltyStatus vs EnrollmentType (Relative counts)
EnrollmentType
2021 Promotion
Standard
LoyaltyStatus vs Province or State (Absolute counts)
LoyaltyStatus vs Province or State (Relative counts)
Province or State
Alberta
British Columbia
Manitoba
New Brunswick
Newfoundland
Nova Scotia
Ontario
Prince Edward Island
Quebec
Saskatchewan
Yukon
Count
0
2000
4000
6000
0.0
0.2
0.4
0.6
0.8
1.0
Aurora
Nova
Star
LoyaltyStatus

Annex2:

New Value Based- General Metrics	Description
Total_NumFlights	Total number of flights.
Total_DistanceKM	Total distance of flights on KM
Perc_Flights_With_Companions	Percentage of flights with companions.
Dollar_Cost_Points_Remaining	Dollar value of points available for redemption.
Recency	Days since last Purchase (if did not make any purchase should be Null).

New Behavioral - Year Agregation	Description
Total_NumFlights_Per_Year	Total number of flights per year (one column per year).
Total_DistanceKM_Per_Year	Total distance traveled per year in KM (one column per year).
Perc_Flights_With_Companions_Per_Year	Percentage of flights with companions per year (one column per year).
Dollar_Cost_Points_Remaining_Per_Year	Dollar value of points remains per year (one column per year).

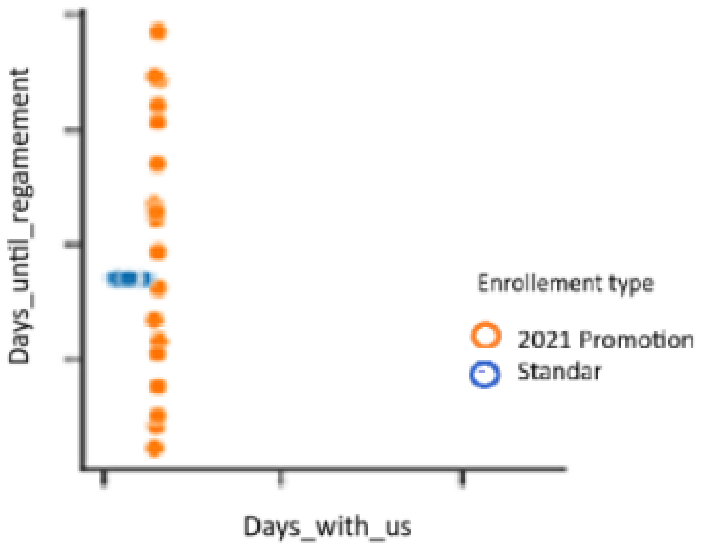
New Behavioral - Month Aggregation	Description
Total_NumFlights_Per_Month	Total number of flights per month (e.g.,Total_NumFlights_month_1).
Total_DistanceKM_Per_Month	Total distance traveled per month on KM.
Perc_Flights_With_Companions_Per_Month	Percentage of flights with companions per month.
Dollar_Cost_Points_Remaining_Per_Month	Dollar value of points remaining per month.

Variables for Clusters	
Based on the work that we will need to perform on the second phase of the project, we have 83 variables, and we will explain in which type of preliminary segmentation they are going to be used:	
Value-Based Segmentation *Groups of customers by their economic value or profitability to the company*	'LoyaltyStatus', 'EnrollmentDateOpening', 'CancellationDate', 'Customer Lifetime Value', 'EnrollmentType', 'Re_engaged_people', 'Days_with_us', 'Days_until_reengagement', 'Total_NumFlights', 'Total_DistanceKM', 'Perc_Flights_With_Companions', 'Dollar_Cost_Points_Remaining', 'Recency_Days'. The ones that probably we will **NOT** use on the cluster are: 'EnrollmentDateOpening', 'CancellationDate', 'Re_engaged_people'
Behavioral Segmentation *Classifies customers based on their actions, habits, and engagement patterns with the brand.*	All the total of flights, total distance, percentage of flights with companions and dollar cost points remaining, per month and per year.

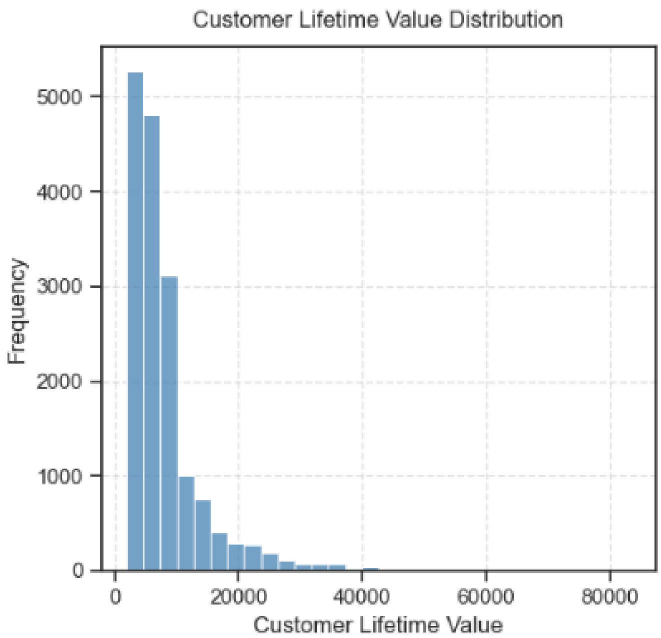
Demographic Segmentation *Divides customers according to personal and social characteristics such as age, gender, income, or location*	'Province or State', 'City', 'Latitude', 'Longitude', 'Postal code', 'Gender', 'Education', 'Location Code', 'Marital Status', 'Income'. <u>The ones that probably we will **NOT** use on the cluster are:</u> 'Province or State', 'City', 'Postal code' and 'Location Code'.

Note: The factor that we used to "remove" variables form the clusters are that they can be obtain using other variables.

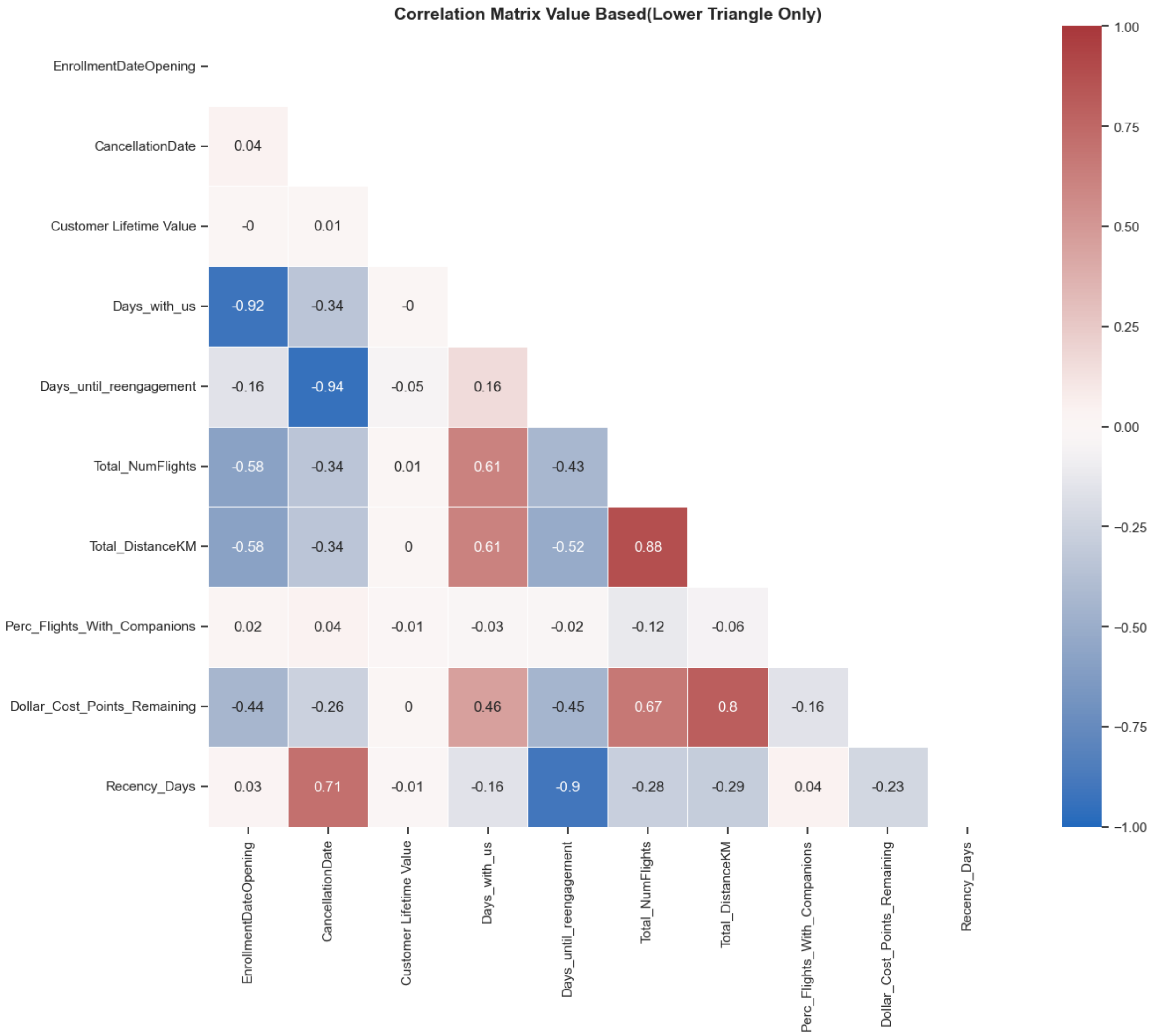
Annex3:



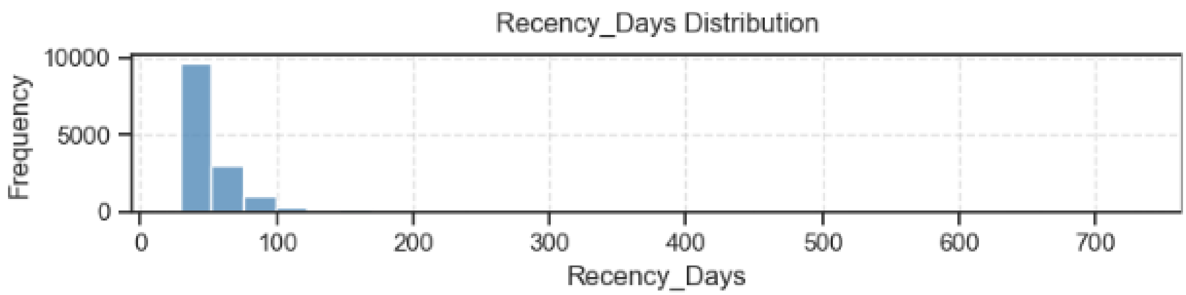
Annex4:



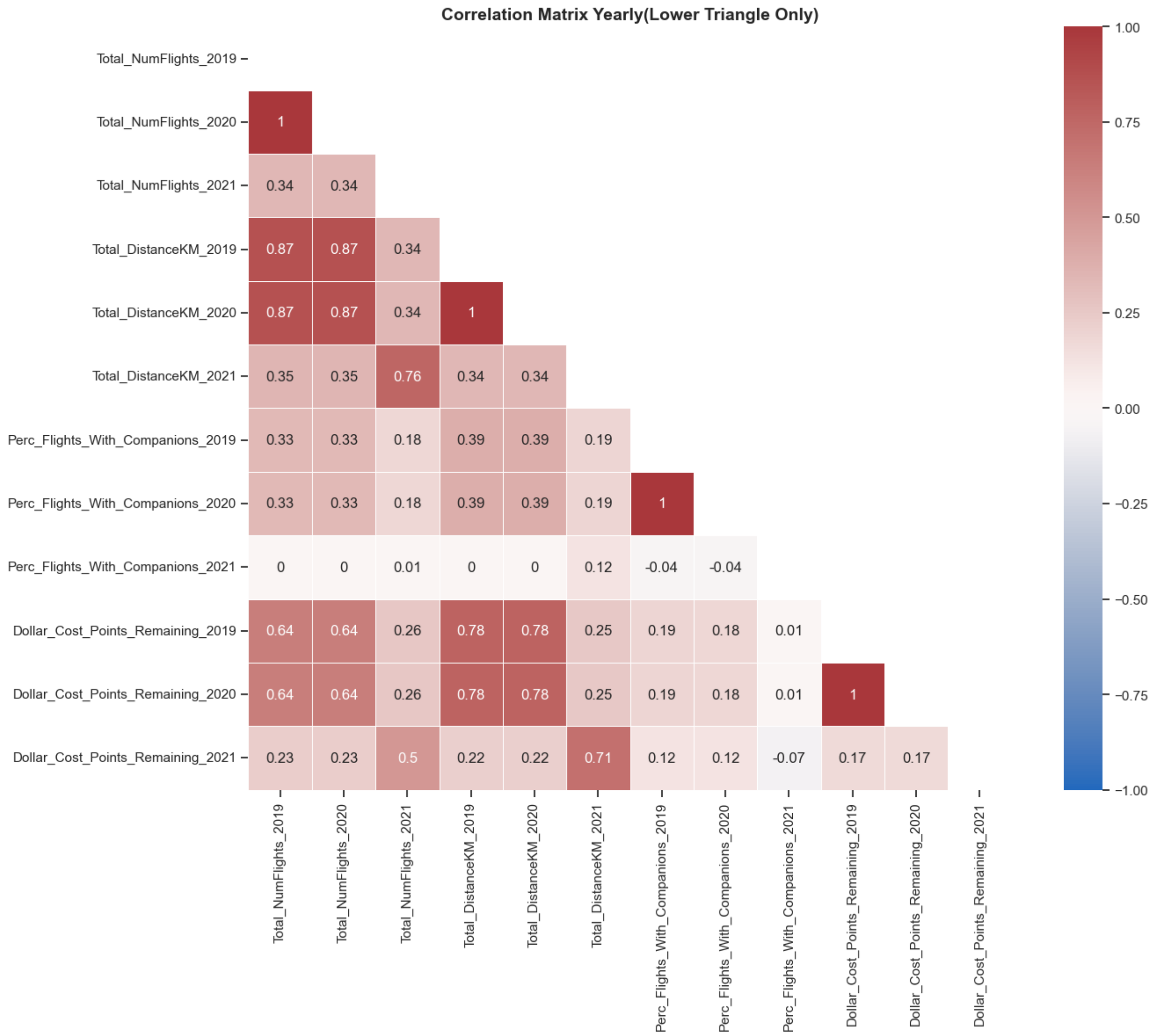
Annex5:



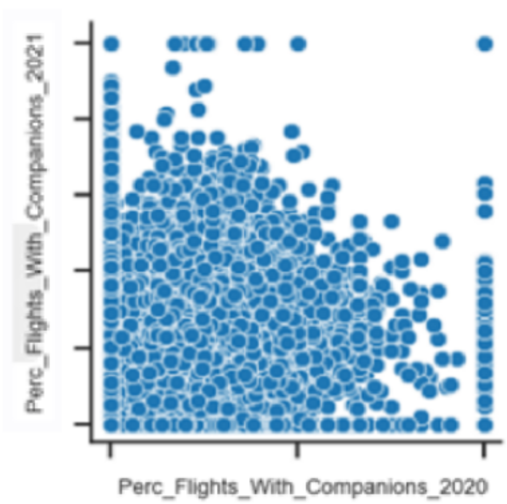
Annex6:



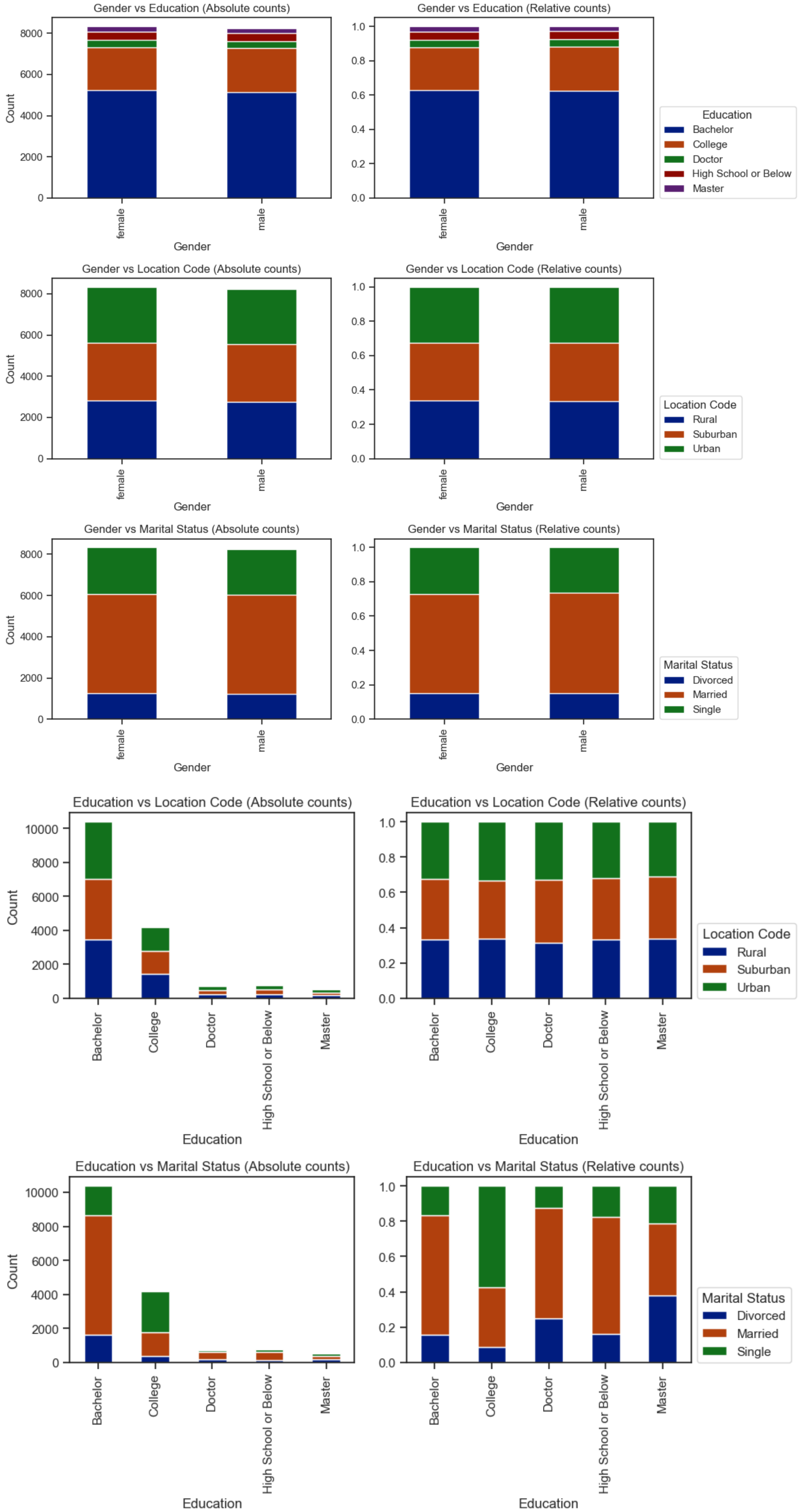
Annex7:



Annex8:

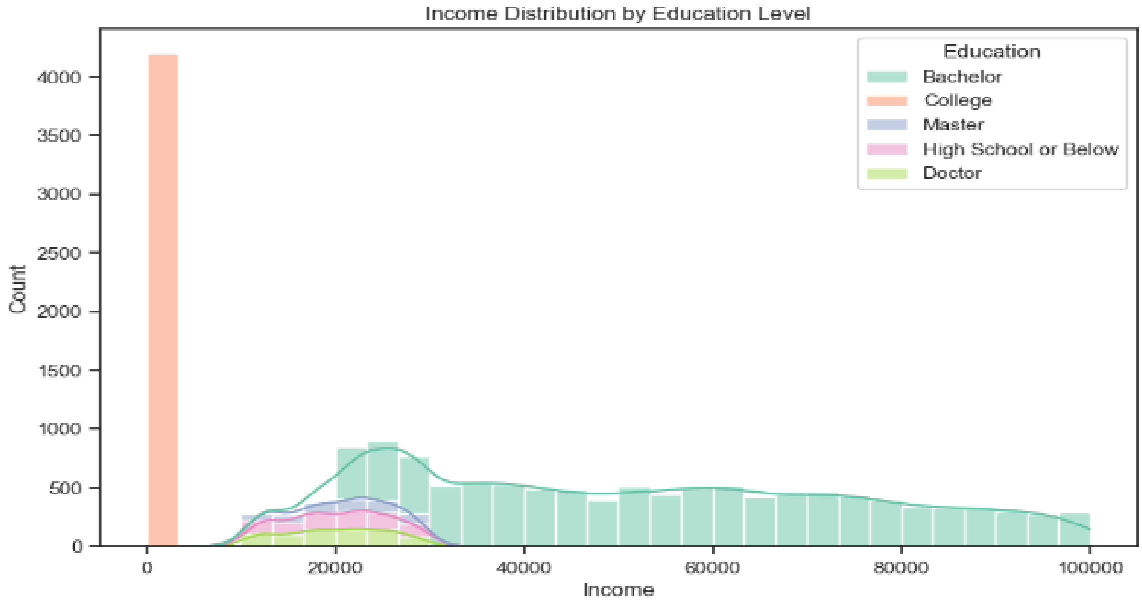


Annex9:

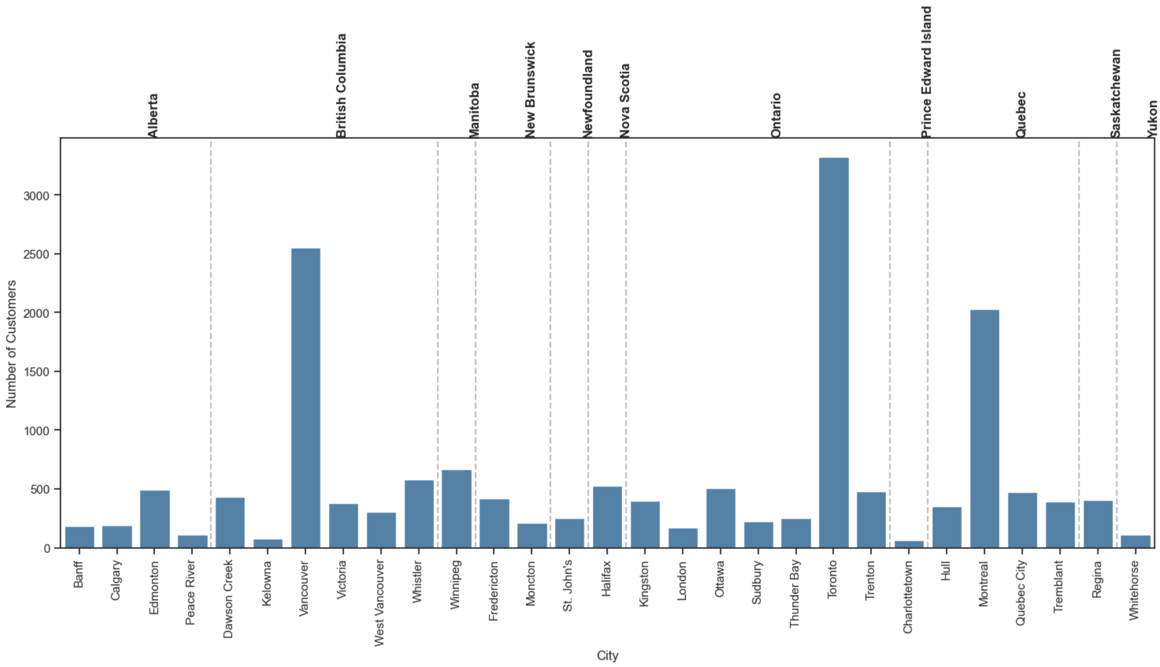


Gender vs Education (Absolute counts)
Gender vs Education (Relative counts)
Education
Bachelor
College
Doctor
High School or Below
Master
Gender vs Location Code (Absolute counts)
Gender vs Location Code (Relative counts)
Location Code
Rural
Suburban
Urban
Gender vs Marital Status (Absolute counts)
Gender vs Marital Status (Relative counts)
Marital Status
Divorced
Married
Single
Count
Gender
female
male
Education vs Location Code (Absolute counts)
Education vs Location Code (Relative counts)
Education vs Marital Status (Absolute counts)
Education vs Marital Status (Relative counts)
Education

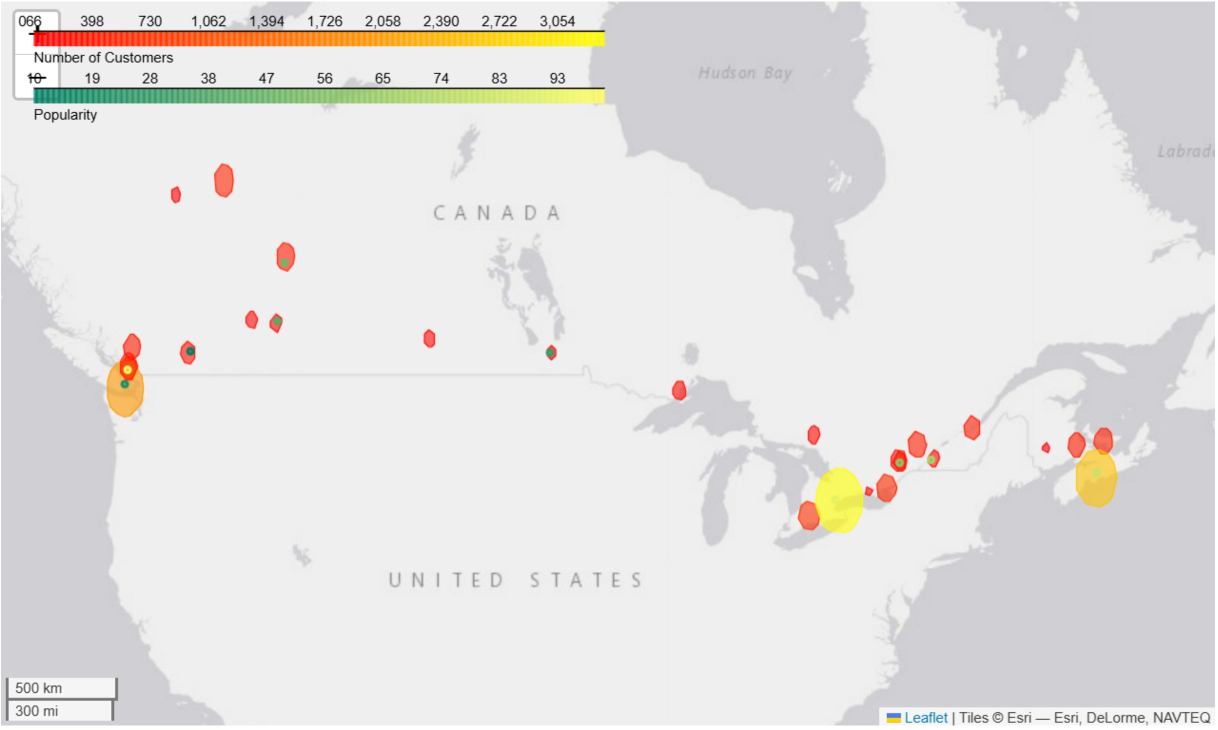
Annex10:



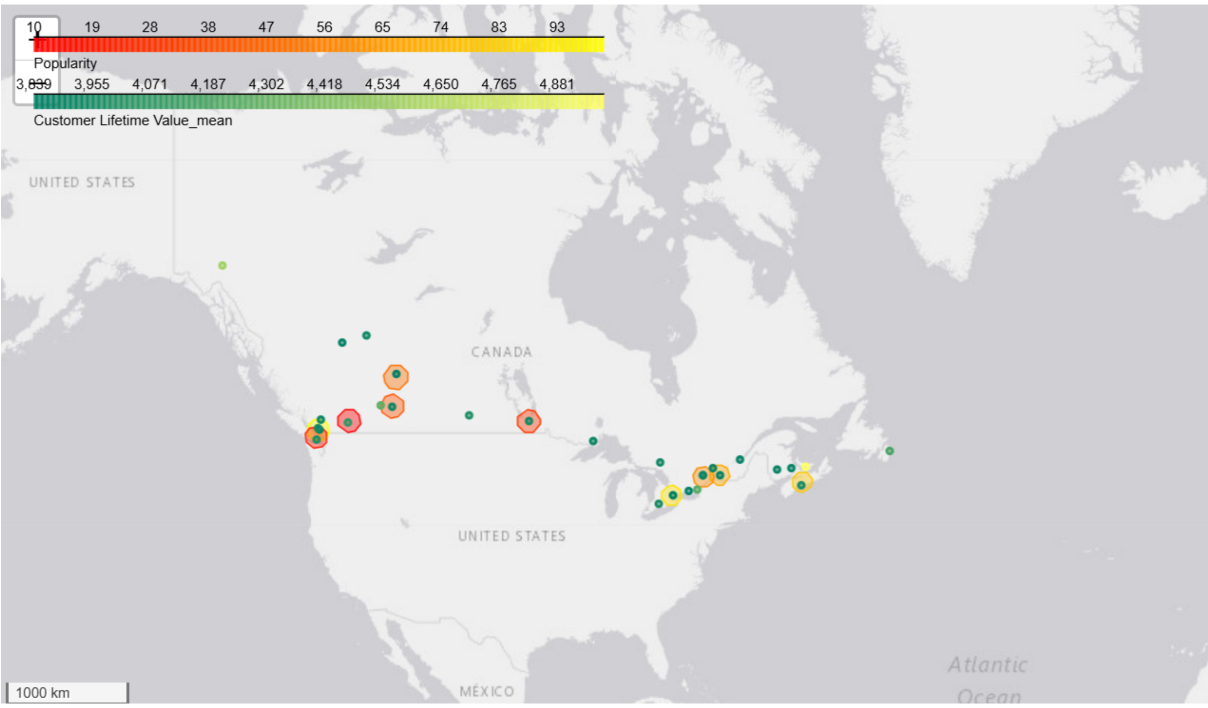
Annex11:



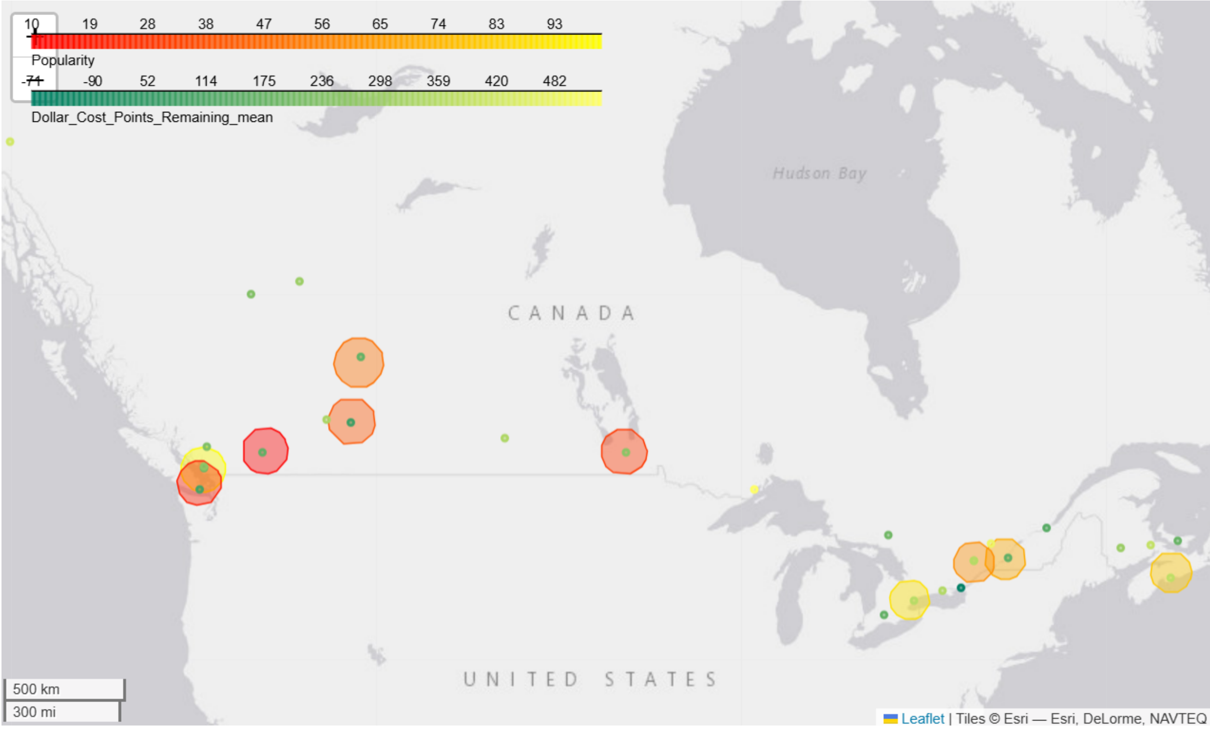
Annex12 – Popular Airports and numbers of customers:



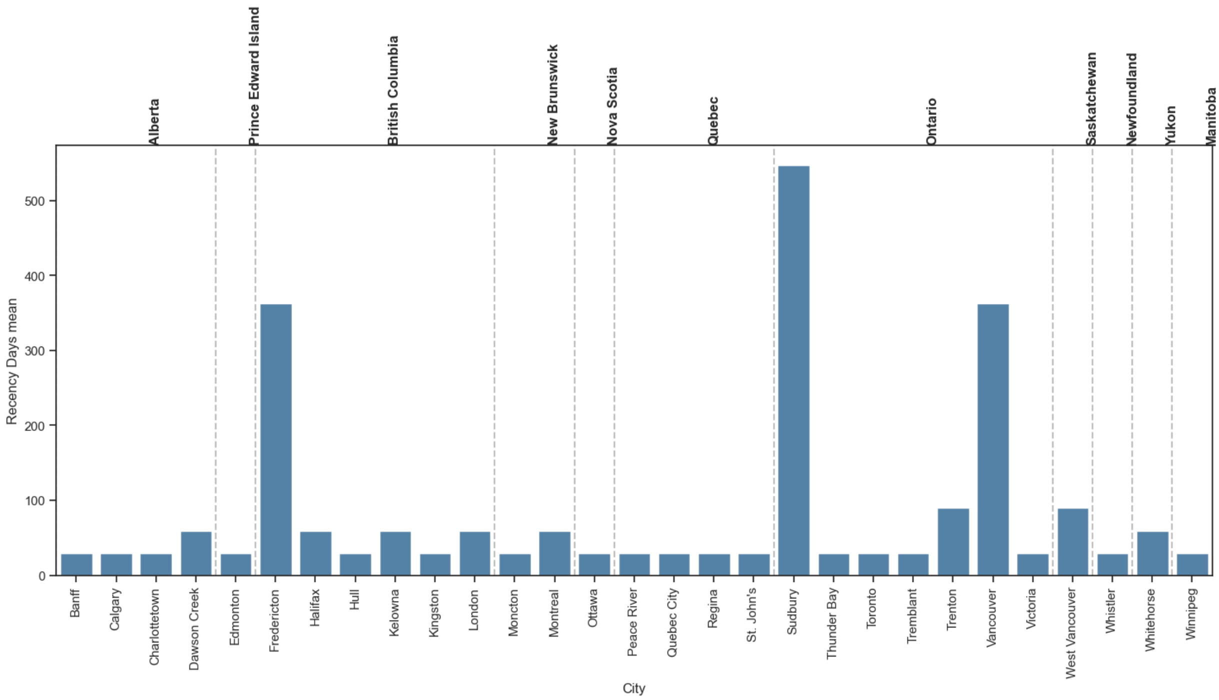
Annex13 - Customer lifetime Value mean:



Annex14 - Points Remaining mean:



Annex15:



BIBLIOGRAPHICAL REFERENCES

Team Digit. (n.d.). *Airports in Canada: List of 21 Domestic & International Airports*. Go Digit General Insurance Limited. https://www.godigit.com/international-travel-insurance/airports/airports-in-canada

APPENDIX

Table of Figures